

Program :

use college

output :

```
switched to db college
```

```
db.createCollection("Student")
```

output :

```
{ "ok" : 1 }
```

```
db.Student.insertMany([
  { roll_no: 1, name: "Amit", class: "TY", division: "A", city: "Pune", TOC: 80, DBMS: 55, SPM: 65 },
  { roll_no: 2, name: "Riya", class: "TY", division: "B", city: "Mumbai", TOC: 70, DBMS: 45, SPM: 75 },
},
  { roll_no: 3, name: "Kiran", class: "TY", division: "B", city: "Pune", TOC: 60, DBMS: 30, SPM: 50 },
  { roll_no: 4, name: "Neha", class: "TY", division: "A", city: "Pune", TOC: 90, DBMS: 65, SPM: 80 },
},
  { roll_no: 5, name: "Sohan", class: "TY", division: "C", city: "Nagpur", TOC: 85, DBMS: 58, SPM: 72 },
},
  { roll_no: 6, name: "Priya", class: "TY", division: "B", city: "Mumbai", TOC: 75, DBMS: 35, SPM: 60 },
},
  { roll_no: 7, name: "Arjun", class: "TY", division: "C", city: "Pune", TOC: 68, DBMS: 40, SPM: 55 },
  { roll_no: 8, name: "Meena", class: "TY", division: "A", city: "Delhi", TOC: 88, DBMS: 62, SPM: 78 },
},
  { roll_no: 9, name: "Vikas", class: "TY", division: "B", city: "Pune", TOC: 72, DBMS: 50, SPM: 69 },
  { roll_no: 10, name: "Sneha", class: "TY", division: "C", city: "Nagpur", TOC: 65, DBMS: 42, SPM:
73 }
]);
```

Output :

```
{
  "acknowledged" : true,
  "insertedIds" : [
    ObjectId("68d3e419cf5c72e26befba3b"),
    ObjectId("68d3e419cf5c72e26befba3c"),
    ObjectId("68d3e419cf5c72e26befba3d"),
    ObjectId("68d3e419cf5c72e26befba3e"),
    ObjectId("68d3e419cf5c72e26befba3f"),
```

```
ObjectId("68d3e419cf5c72e26befba40"),
ObjectId("68d3e419cf5c72e26befba41"),
ObjectId("68d3e419cf5c72e26befba42"),
ObjectId("68d3e419cf5c72e26befba43"),
ObjectId("68d3e419cf5c72e26befba44")
]
}
```

1. Find average of total marks in toc

```
db.Student.aggregate([
  { $group: { _id: null, Avg_TOC: { $avg: "$TOC" } } }
])
```

output :

```
{ "_id" : null, "Avg_TOC" : 75.3 }
```

2. Find students of div b who scored min marks in DBMS (partial)

```
db.Student.aggregate([
  { $match: { division: "B" } },
  { $group: { _id: null, minDBMS: { $min: "$DBMS" } } }
])
```

output :

```
{ "_id" : null, "minDBMS" : 30 }
```

3. Find total sum of marks in SEPM of students staying in pune

```
db.Student.aggregate([
  { $match: { city: "Pune" } },
  { $group: { _id: null, Total_SEPM: { $sum: "$SPM" } } }
])
```

output :

```
{ "_id" : null, "Total_SPM" : 319 }
```

4. Find the division wise count of students whose dbms marks is greater than 30 and less than 60

```
db.Student.aggregate([
  { $match: { DBMS: { $gt: 30, $lt: 60 } } },
```

```
{ $group: { _id: "$division", Count: { $sum: 1 } } }
})
```

output :

```
{ "_id" : "A", "Count" : 1 }
{ "_id" : "B", "Count" : 3 }
{ "_id" : "C", "Count" : 3 }
```

5.Find the count of students from each city

```
db.Student.aggregate([
  { $group: { _id: "$city", Count: { $sum: 1 } } }
])
```

output :

```
{ "_id" : "Delhi", "Count" : 1 }
{ "_id" : "Pune", "Count" : 5 }
{ "_id" : "Mumbai", "Count" : 2 }
{ "_id" : "Nagpur", "Count" : 2 }
```

6.Find the maximum marks of student in sepm who stay in the same city

```
db.Student.aggregate([
  { $group: { _id: "$city", Max_SEPM: { $max: "$SEPM" } } }
])
```

output :

```
{ "_id" : "Delhi", "Max_SEPM" : 78 }
{ "_id" : "Pune", "Max_SEPM" : 80 }
{ "_id" : "Mumbai", "Max_SEPM" : 75 }
{ "_id" : "Nagpur", "Max_SEPM" : 73 }
```

7.Create compound index on name and class

```
db.Student.createIndex({ name: 1, class: 1 })
```

output :

```
{
  "numIndexesBefore" : 1,
  "numIndexesAfter" : 2,
```

```
"createdCollectionAutomatically" : false,  
"ok" : 1  
}
```

8.Create single index on name

```
db.Student.createIndex({ name: 1 })
```

output:

```
{  
"numIndexesBefore" : 2,  
"numIndexesAfter" : 3,  
"createdCollectionAutomatically" : false,  
"ok" : 1  
}
```